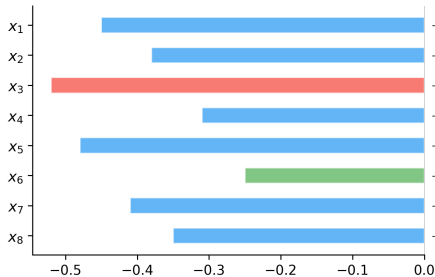
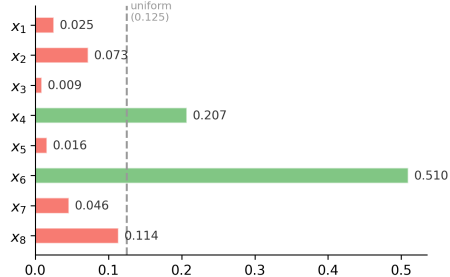


Step 0 — Initialize: sample N particles from frozen generative model $\{x_i\}_{i=1}^N \sim p_\theta(x)$

Step 1: Score all particles



Step 2: Reweight $w_i \propto \exp(\Delta\beta \cdot s_i)$, $\Delta\beta = 15$



x₆ (score -0.25, best) → weight 51.0% — 4.1× more likely to survive than uniform (12.5%)

Step 5: Mutate (mask → predict → fill)

1. Before (duplicated):

A C G T A T G C A C G T

2. Mask 3/12 positions:

A C ? T A T G ? A C ? T



GenModel fills

3. After (new particle):

A C T T A T G A A C A T

Changes: pos 3: G→T, pos 8: C→A, pos 11: G→A

Why generative > random

Model sees UNMASKED positions as context. Predictions respect DNA grammar: motifs, GC balance, etc.

Restores diversity

All N duplicates get INDEPENDENT random masks → diverge into N distinct new sequences

n_f controls exploration

Small n_f (0.05) = local search
Large n_f (0.20) = big jumps
~10 positions/step for 200bp